TODAY'S TOPICS 16/10/26

- RADIAL DISTRIBUTION FUNCTION (RDF) $g(r)$
- ENERGY AND PRESSURE IN TERMS OF RDF
- TAIL CORRECTIONS

RADIAL DISTRIBUTION FUNCTION

$\tilde{\rho}(\vec{r}_1, \vec{r}_2)$ = { PROBABILITY OF FINDING A PARTICLE IN $\vec{r}_1$ AND ANOTHER ONE IN $\vec{r}_2$ }

EVENT A = { A PARTICLE IS IN $\vec{r}_1$ }

EVENT B = { // // // // $\vec{r}_2$ }

$$P(A \cap B) = \tilde{\rho}(\vec{r}_1, \vec{r}_2)$$

$$P(A \cap B) = P(A|B) \cdot P(B)$$

↑ CONDITIONAL PROBABILITY

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

NOTE THAT (HOMOGENEOUS CASE):
IF $P(A) = P(B)$
$P(A|B) = \frac{P(A \cap B)}{P(B)} =$
$= \frac{P(A \cap B)}{P(A)} = P(B|A)$

IF A AND B INDEPENDENT

$$P(A|B) = \frac{P(A)\,\cancel{P(B)}}{\cancel{P(B)}} = P(A)$$

GOING BACK TO $\tilde{\rho}$:

$$\tilde{\rho}(\vec{r}_1, \vec{r}_2) = \tilde{\rho}(\vec{r}_1|\vec{r}_2)\,\tilde{\rho}(\vec{r}_2) \qquad (*)$$

WE DEFINE:

$$g^{(2)}(\vec{r}_1, \vec{r}_2) \equiv \frac{\tilde{\rho}(\vec{r}_1, \vec{r}_2)}{\tilde{\rho}(\vec{r}_1)\,\tilde{\rho}_2(\vec{r}_2)}$$

} THIS IS 2-PARTICLE (PAIR) DISTRIBUTION FUNCTION

IF THE SYSTEM IS HOMOGENEOUS $\tilde{\rho}(\vec{r}_1) = \tilde{\rho}(\vec{r}_2) = \tilde{\rho}$

SO THAT:

$$g^{(2)}(\vec{r}_1, \vec{r}_2) = \frac{\tilde{\rho}(\vec{r}_1, \vec{r}_2)}{\tilde{\rho}^2}$$

BUT USING (*) WE HAVE:

$$g^{(2)}(\vec{r}_1, \vec{r}_2) = \frac{\tilde{\rho}(\vec{r}_1|\vec{r}_2)\,\cancel{\tilde{\rho}}}{\tilde{\rho}\,\cancel{\tilde{\rho}}}$$

$$= \frac{\tilde{\rho}(\vec{r}_1 \mid \vec{r}_2)}{\tilde{\rho}}$$

SINCE SYSTEM IS HOMOGENEOUS WE CAN EXPECT

$g^{(2)}(\vec{r}_1, \vec{r}_2) = g^{(2)}(\vec{r}_1 - \vec{r}_2)$

SO THAT $g^{(2)}(\vec{r}_1 - \vec{r}_2) = g^{(2)}(\vec{r})$

AND $\tilde{\rho}(\vec{r}_1 \mid \vec{r}_2) = \tilde{\rho}(\vec{r})$

I.E. $\tilde{\rho}\, g^{(2)}(\vec{r}_1, \vec{r}_2) = \tilde{\rho}(\vec{r}_1 \mid \vec{r}_2)$

IDEAL GAS

$$\tilde{\rho}(\vec{r}_1 \mid \vec{r}_2) = \tilde{\rho}(\vec{r}_1) = \tilde{\rho}$$

HENCE

$$g^{(2)}(\vec{r}_1, \vec{r}_2) = \frac{\tilde{\rho}}{\tilde{\rho}} = 1$$

HARD SPHERES

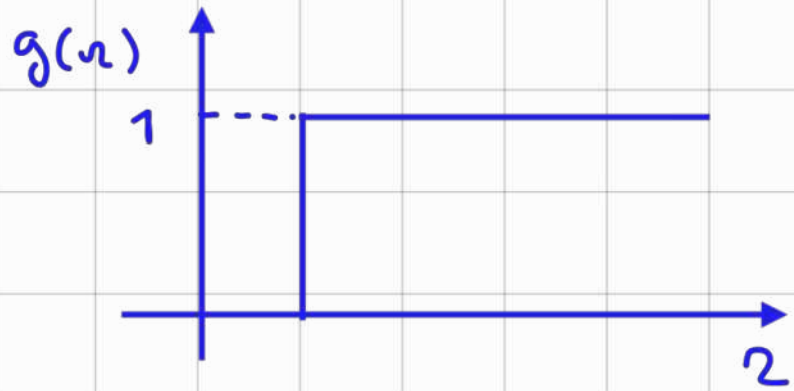
IF $\|\vec{r}_1 - \vec{r}_2\| < \sigma \Rightarrow \rho^{(2)}(\vec{r}_1 \mid \vec{r}_2) = 0$

AND $g^{(2)}(\vec{r}_1 \mid \vec{r}_2) = 0$

MOREOVER SYSTEM IS ISOTROPIC THEREFORE

$g^{(2)}(\vec{r}_1 \mid \vec{r}_2) = g^{(2)}(\|\vec{r}_{12}\|) = g^{(2)}(r)$

RADIAL DISTRIBUTION FUNCTION



AT VERY LOW DENSITIES

BUT AT HIGH DENSITIES ?

WE WILL SEE ...

TWO POINTS DENSITY

IF $\mathcal{N}(\vec{r}_1 \dots \vec{r}_N) = \dfrac{e^{-\beta U_N}}{Z}$

$\mathcal{N}(\vec{r}_1 \dots \vec{r}_N)\, d\vec{r}_1 \dots d\vec{r}_N$ IS THE PROBABILITY OF FINDING PARTICLES AT POSITIONS $\vec{r}_1, \dots \vec{r}_N$

HENCE THE PROBABILITY DENSITY TO FIND ONE PARTICLE IN $\vec{r}_1$ IS:

$$\tilde{\rho}(\vec{r}_1) = \int \frac{e^{-\beta U_N}}{Z}\, d\vec{r}_2 \dots d\vec{r}_N$$

NOTE THAT WE ARE NOT INTEGRATING OVER $d\vec{r}_1$

AND THE (1-PARTICLE) DENSITY IS:

$$\rho(\vec{r}_1) = N \cdot \tilde{\rho}(\vec{r}_1) = N \cdot \int \frac{e^{-\beta U_N}}{Z}\, d\vec{r}_2 \dots d\vec{r}_N$$

N: TOTAL NUMBER OF PARTICLES

NOTE THAT $\int \tilde{\rho}(\vec{r}_1)\, d\vec{r}_1 = 1$ AND THAT $N \tilde{\rho}(\vec{r}_1)\, d\vec{r} = \{$ NUMBER OF PARTICES IN $d\vec{r}_1 \}$

MOREOVER IF THE SYSTEM IS HOMOGENEOUS

$$\rho(\vec{r}_1) = \frac{N}{V}$$

SIMILARLY THE PROBABILITY DENSITY OF FINDING A PARTICLE IN $\vec{r}_1$ AND $\vec{r}_2$ IS

$$\tilde{\rho}(\vec{r}_1, \vec{r}_2) = \int \frac{e^{-\beta U_N}}{Z}\, d\vec{r}_3 \dots d\vec{r}_N$$

NOTE THAT WE ARE NOT INTEGRATING OVER $\vec{r}_1$ AND $\vec{r}_2$

SO THAT

$\underbrace{N(N-1)}_{\downarrow}\ \tilde{\rho}(\vec{r}_1,\vec{r}_2) = \{$ NUMBER OF PAIRS IN $d\vec{r}_1$ AND $d\vec{r}_2$ $\}$

TOTAL NUMBER OF POSSIBLE PAIRS

AND WE DEFINE THE TWO PARTICLE DENSITY AS FOLLOWS

$$\rho^{(2)}(\vec{r}_1,\vec{r}_2) = N(N-1)\int \frac{e^{-\beta U_N}}{Z} d^{(N-2)}\vec{r}$$

IDEAL GAS

$$\rho^{(2)}(\vec{r}_1,\vec{r}_2) = N(N-1)\int \frac{e^{-\beta U_N}}{Z}\, d\vec{r}_3 \ldots d\vec{r}_N$$

$U_N = 0$

$$= N(N-1)\int \frac{d\vec{r}_3 \ldots d\vec{r}_N}{Z} = \overbrace{N(N-1)}^{\approx N^2}\frac{V^{N-2}}{V^N} =$$

SINCE $Z = \int e^{-\beta U_N} d\vec{r}_1 \ldots d\vec{r}_N = V^N$

$$\approx \frac{N^2}{V^2} = \rho^2 \quad \text{HENCE}$$

$$g^{(2)}(\vec{r}_1,\vec{r}_2) = \frac{\rho^{(2)}(\vec{r}_1,\vec{r}_2)}{\rho^2} = \frac{\rho^2}{\rho^2} = 1$$

ALTERNATIVE WAYS OF CALCULATING 1-POINT AND 2-POINT DENSITIES

$U_N = U_N(\vec{r}_1,\vec{r}_2 \ldots \vec{r}_N)$

$$\rho(\vec{r}) = \frac{N}{Z}\int e^{-\beta U_N} d^{(N-1)}\vec{r} =$$

NOW $U_N = U_N(\vec{r}_1 \ldots \vec{r}_N)$

$$= \frac{N}{Z}\int e^{-\beta U_N}\, \delta(\vec{r}-\vec{r}_1)\, d^N\vec{r} =$$

INTEGRAL DOES NOT DEPEND ON PARTICLE INDEX

$$= \frac{1}{N}\sum_i^N \int e^{-\beta U_N}\, \delta(\vec{r}-\vec{r}_i)\, d^N\vec{r}$$

TAKE A SYSTEM OF 3 PARTICLES CONSIDER $\vec{r}_2$ AND $\vec{r}_3$ INSTEAD OF $\vec{r}_1$, THEN SWAP INDICES TO RECOVER CASE ABOVE ($\vec{r}_1$)

$$= \left\langle \sum_i^N \delta(\vec{r}-\vec{r}_i) \right\rangle$$

NOTE THAT $U_2(\vec{r}_{iT}) = U_2(\vec{r}_{Ti})$

WHERE $\langle \dots \rangle = \int \dots \frac{e^{-\beta U_N}}{Z} d^N\vec{r}$

SIMILARLY

$$\rho^{(2)}(\vec{r}', \vec{r}) = \left\langle \sum_{i,j}{}' \delta(\vec{r}' - \vec{r}_i)\, \delta(\vec{r} - \vec{r}_j) \right\rangle$$

($'$: $i \neq j$)

INDEED

$$\rho^{(2)}(\vec{r}', \vec{r}) = N(N-1) \int \frac{e^{-\beta U_N}}{Z} d^{(N-2)}\vec{r} =$$

($U_N = U_N(\vec{r}', \vec{r}, \vec{r}_3 \dots \vec{r}_N)$; $U_N = U_N(\vec{r}_1 \dots \vec{r}_N)$)

$$= N(N-1) \int \delta(\vec{r}' - \vec{r}_1)\, \delta(\vec{r} - \vec{r}_2) \frac{e^{-\beta U_N}}{Z} d\vec{r}_1 d\vec{r}_2 \cdot d^{(N-2)}\vec{r}$$

TAKE AGAIN A SYSTEM OF 3 PARTICLES AND USE $(\vec{r}_2, \vec{r}_3)$ INSTEAD OF $(\vec{r}_1, \vec{r}_2)$

INTEGRAL DOES NOT DEPEND ON WHICH PAIR OF INDICES WE ARE CONSIDERING

$$= \sum_{i,j}{}' \int \delta(\vec{r}' - \vec{r}_i)\, \delta(\vec{r} - \vec{r}_j) \frac{e^{-\beta U_N}}{Z} d^N\vec{r} =$$

$$= \left\langle \sum_{ij}{}' \delta(\vec{r}' - \vec{r}_i)\, \delta(\vec{r} - \vec{r}_j) \right\rangle = \rho^{(2)}(\vec{r}', \vec{r})$$

$\rho^{(2)}(\vec{r}', \vec{r})$ IN THE HOMOGENEOUS CASE

LET'S CONSIDER

$$\left\langle \frac{1}{N\rho} \sum_{i,j}{}' \delta(\vec{r} - (\vec{r}_i - \vec{r}_j)) \right\rangle =$$

($\vec{r}_{ij}$; FUNCTION OF $\vec{r}$ AND WE AVERAGE OVER $\vec{r}_1 \dots \vec{r}_N$)

$$= \left\langle \frac{1}{N\rho} \sum_{i,j}{}' \delta(\vec{r} - \vec{r}_i + \vec{r}_j) \right\rangle =$$

$$= \left\langle \frac{1}{N\rho} \int \sum_{i,j}{}' \delta(\vec{r} - \vec{r}_i + \vec{r}') \cdot \delta(\vec{r}' - \vec{r}_j)\, d\vec{r}' \right\rangle =$$

($\rho^{(2)}(\vec{r} + \vec{r}', \vec{r}')$)

$$= \frac{1}{N\rho} \int \left\langle \sum_{i,j}{}' \delta((\vec{r} + \vec{r}') - \vec{r}_i) \cdot \delta(\vec{r}' - \vec{r}_j) \right\rangle d\vec{r}' =$$

$$= \frac{1}{N\rho} \int \rho^{(2)}(\vec{r}+\vec{r}', \vec{r}')\, d\vec{r}' =$$

BUT THE SYSTEM IS HOMOGENEOUS

THEREFORE $\rho^{(2)}(\vec{r}+\vec{r}', \vec{r}') = \rho^{(1)}(\vec{r})$

I.E IT IS INDEPENDENT OF $\vec{r}'$

$$= \frac{1}{N\rho} \int \rho^{(1)}(\vec{r})\, d\vec{r}' = \frac{V}{N\rho}\, \rho^{(1)}(\vec{r})$$

$$= \frac{1}{\rho^2}\, \rho^{(2)}(\vec{r}) = \frac{g^{(2)}(\vec{r})\cdot \cancel{\rho^2}}{\cancel{\rho^2}} =$$

$$= g^{(2)}(\vec{r})$$

WE PROVED THAT

$$g^{(2)}(\vec{r}) = \frac{1}{N\rho} \left\langle \sum_{ij}{}' \delta(\vec{r} - (\vec{r}_i - \vec{r}_j)) \right\rangle$$

$$= \frac{1}{N\rho} \left\langle \sum_{ij}{}' \delta(\vec{r} - \vec{r}_{ij}) \right\rangle \qquad \vec{r}'_{ij} = \vec{r}_i - \vec{r}_j$$

AVERAGE VALUE OF $g(\vec{r})$ OVER A VOLUME ΔV AROUND $\vec{r}$

$$\bar{g}(\vec{r}) = \frac{1}{\Delta V} \int_{\Delta V_{\vec{r}}} g(\vec{r}')\, d\vec{r}'$$

$\bar{g}(\vec{r})$: DEPENDENT ON CHOSEN $\Delta V_{\vec{r}}$ (IF $\Delta V_{\vec{r}}$ IS A CUBIC BOX, $\bar{g}$ DEPENDS ON ITS POSITION $\vec{r}$)

$\Delta V_{\vec{r}}$: THIS IS TO REMIND THAT VOLUME ΔV IS AROUND $\vec{r}$

IN A MC SIMULATION:

$$\langle O(\vec{r}^N) \rangle = \frac{1}{L} \sum_{k=1}^{L} O(r_k^N) = \frac{1}{L} \sum_{k=1}^{L} O_k$$

L: OR EVERY A CERTAIN AMOUNT OF STEPS

$\sum_{k=1}^{L}$: SUM OVER MC STEPS

HENCE CONSIDER THE FOLLOWING OBSERVABLE

$$\bar{g}(\vec{r}; \vec{r}_1 \ldots \vec{r}_N) \equiv \frac{1}{\Delta V} \int_{\Delta V_{\vec{r}}} \left[\frac{1}{N\rho} \sum_{ij}{}' \delta(\vec{r}' - \vec{r}_{ij})\, d\vec{r}' \right]$$

$g(\vec{r})$ OF ONE CONFIGURATION (NO AVERAGE OVER $\frac{e^{-\beta U}}{Z}$)

ITS AVERAGE VALUE IS

$$\langle \bar{g} \rangle = \left\langle \frac{1}{\Delta V} \int_{\Delta V_{\vec{r}}} \frac{1}{N\rho} \sum_{iT}{}' \delta(\vec{r}' - \vec{r}_{iT})\, d\vec{r}' \right\rangle$$

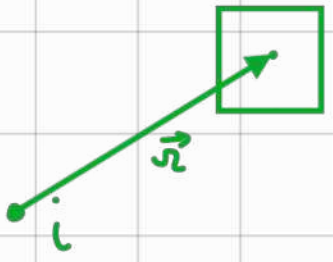
$$= \int \frac{1}{\Delta V} \int_{\Delta V_{\vec{r}}} \frac{1}{N\rho} \sum_{iT}{}' \delta(\vec{r}' - \vec{r}_{iT})\, d\vec{r}' \frac{e^{-\beta U_N}}{Z}\, d\vec{r}_1 \ldots d\vec{r}_N d\vec{r}$$

$$= \frac{1}{L} \sum_{k=1}^{L} \underbrace{\frac{1}{\Delta V} \int_{\Delta V_{\vec{r}}} \frac{1}{N\rho} \sum_{i,T}{}' \delta(\vec{r}' - \vec{r}_{iT}^{(k)})\, d\vec{r}'}_{\bar{g}_k} = \frac{1}{L} \sum_{k}^{L} \bar{g}_k$$

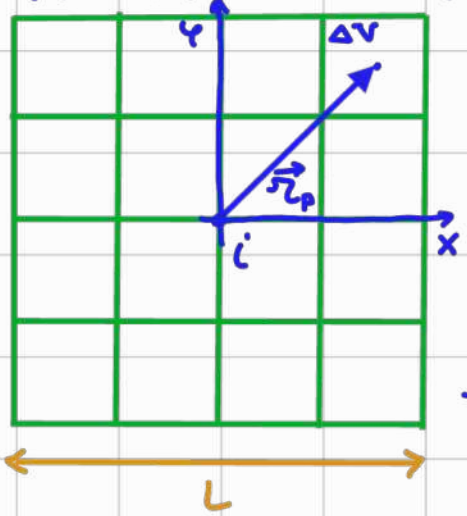
$\bar{g}_k$ IS OUR OBSERVABLE (AS $\mathcal{O}_k$) CALCULATED AT k-th MC STEP

$$\bar{g}_k(\vec{r}) = \frac{1}{\Delta V} \int_{\Delta V_{\vec{r}}} \frac{1}{N\rho} \sum_{i,T}{}' \delta(\vec{r}' - \vec{r}_{iT}^{(k)})\, d\vec{r}'$$

$$= \overbrace{\frac{1}{\Delta V \rho} \frac{1}{N} \sum_{i}}^{\text{AVERAGE OVER ALL PARTICLES}} \underbrace{\int_{\Delta V_{\vec{r}}} \sum_{T}{}' \delta(\vec{r}' - \vec{r}_{iT}^{(k)})\, d\vec{r}'}_{\#\text{ PARTICLES IN } \Delta V_{\vec{r}} \text{ A DISTANCE } \vec{r} \text{ FROM PARTICLE } i}$$



BOX CENTERED AT i OF VOLUME V DIVIDED INTO M SMALL BOXES OF VOLUME ΔV



$\vec{r}_p$ WITH $p = 1 \ldots M$ IS THE CENTER OF p-th BOX OF VOLUME ΔV

$\vec{r}_p = \Delta r\,(k\hat{x} + \ell\hat{y} + m\hat{z})$

k, ℓ, m INTEGERS

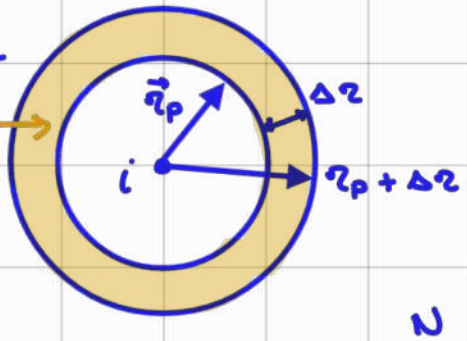
THUS OBTAINING

$\bar{g}_k(\vec{r}_p) \approx g_k(\vec{r}_p)$ IF ΔV SMALL ENOUGH

ISOTROPIC CASE

SPHERICAL SHELL OF THICKNESS Δr

$$\Delta V_p = 4\pi r_p^2 \Delta r$$

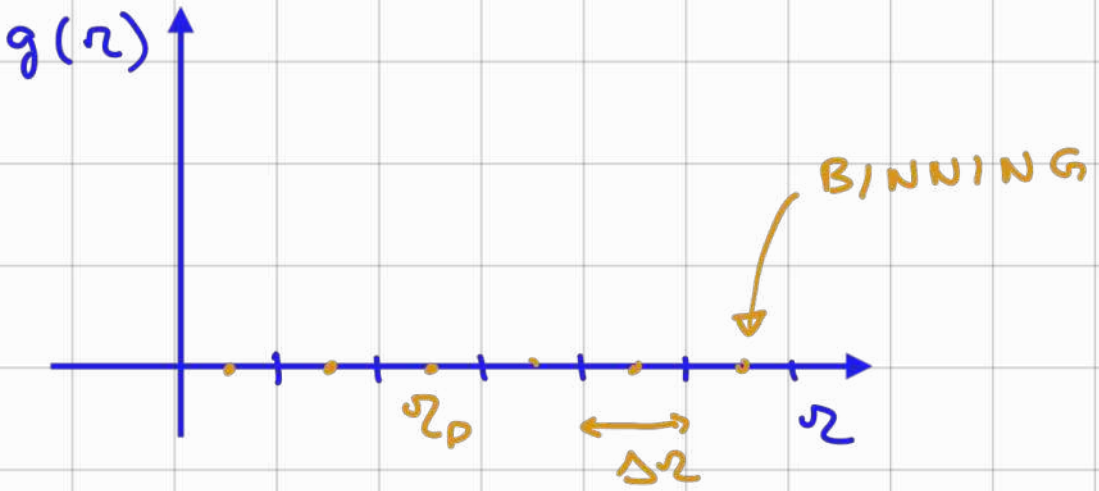


$$\overline{g}_k(r_p) = \underbrace{\frac{1}{4\pi r_p^2 \Delta r_p}}_{\text{AVERAGE NUMBER OF PARTICLES INSIDE } \Delta V_p} \frac{1}{N} \sum_i^N \int_{\Delta V_p} \overbrace{\sum_j{}' \delta(\vec{r} - \vec{r}_{ij})}^{\#\text{ PARTICLE INSIDE p-th SHELL}} d\vec{r} =$$

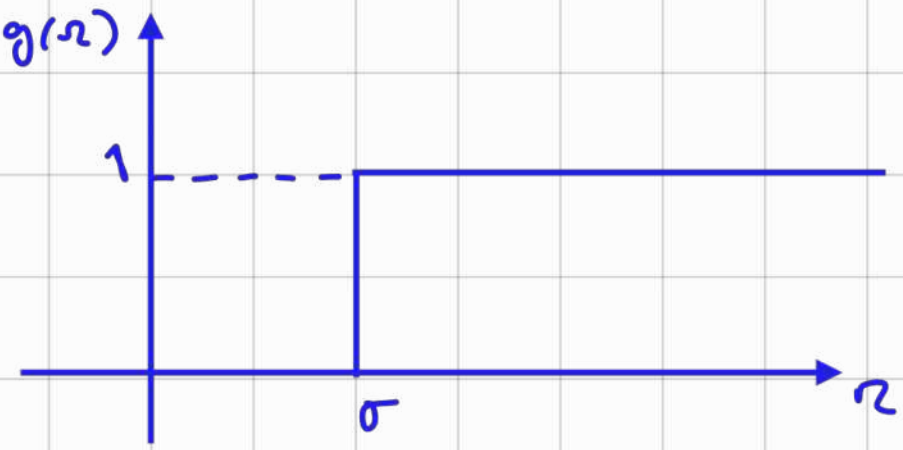
NOTE THAT THIS IS A SCALAR

$\overline{g}_k(r_p) \approx g_k(r_p)$ IS THE RADIAL DISTRIBUTION FUNCTION

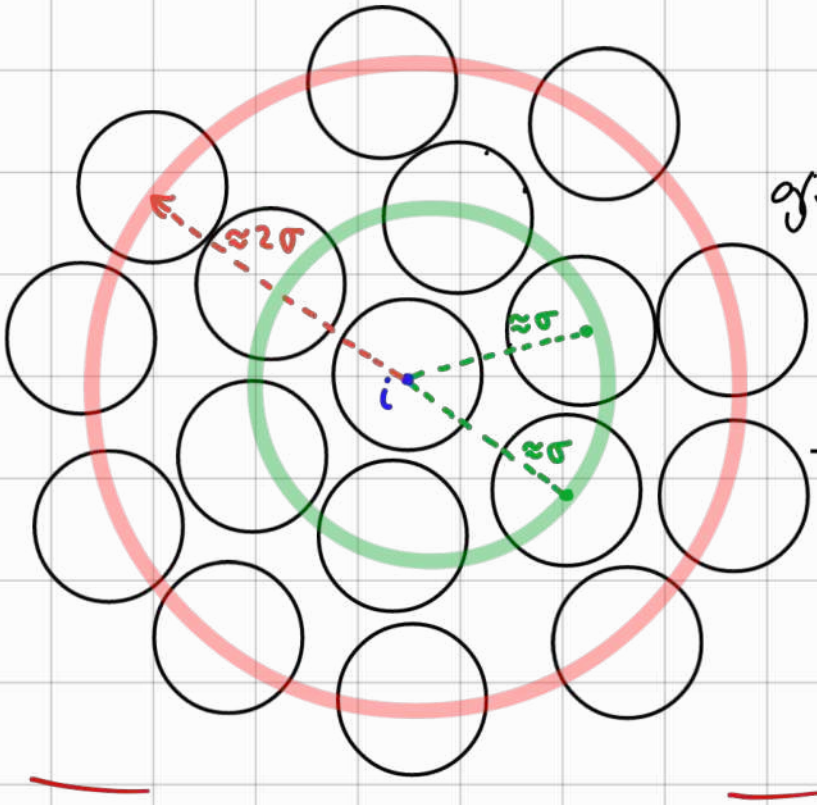
IF Δr IS SMALL ENOUGH

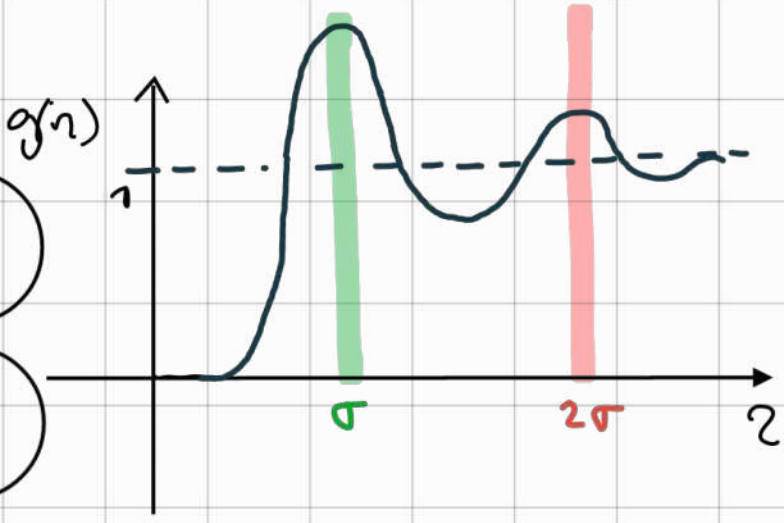


LOW CONCENTRATIONS



HIGH CONCENTRATION IN FLUID PHASE





CALCULATION OF ENERGY FROM g(r)

$$\langle U_N \rangle = U_{ex} = \int \frac{1}{2} \sum_{ij}{}' U_2(\vec{r}_{ij}) \frac{e^{-\beta U_N}}{Z} d^N\vec{r} =$$

$$= \frac{1}{2} N(N-1) \int U_2(\vec{r}_{12}) \frac{e^{-\beta U_N}}{Z} d\vec{r}_1 d\vec{r}_2 d^{(N-2)}\vec{r}$$

$$= \frac{1}{2} \int U_2(\vec{r}_{12}) \underbrace{\int N(N-1) \frac{e^{-\beta U_N}}{Z} d^{(N-2)}\vec{r}}_{\rho^{(2)}(\vec{r}_1, \vec{r}_2)} d\vec{r}_1 d\vec{r}_2$$

$$= \frac{1}{2} \int U_2(\vec{r}_{12}) \rho^{(2)}(\vec{r}_1, \vec{r}_2) d\vec{r}_1 d\vec{r}_2$$

BUT $\rho^{(2)}(\vec{r}_1, \vec{r}_2) = \rho^2 g(\vec{r}_{12})$ ← HOMOGENEOUS CASE

$$= \frac{\rho^2}{2} \int U_2(\vec{r}_{12}) g(\vec{r}_{12}) d\vec{r}_1 d\vec{r}_2$$

IF WE CHANGE VARIABLES

$(\vec{r}_1, \vec{r}_2) \to (\vec{r}_2, \vec{r})$ WHERE

$\vec{r} = \vec{r}_{12} = \vec{r}_1 - \vec{r}_2$

$$= \frac{\rho^2}{2} \int v_2(\vec{r})\, g(\vec{r})\, d\vec{r}\, d\vec{r}_2 =$$

$$= V \frac{\rho^2}{2} \int v_2(\vec{r})\, g(\vec{r})\, d\vec{r}$$

IF THE SYSTEM IS ISOTROPIC

$$= V \frac{\rho^2}{2} \int v_2(r)\, g(r)\, 4\pi r^2\, dr$$

↑ RADIAL DISTRIBUTION FUNCTION

TAIL CORRECTIONS

IN A SIMULATION $v_{ij}(r) \to v_{ij}^c(r)$

WHERE $v_{ij}^c = \begin{cases} v_{ij}(r) & r \le r_c \\ 0 & r > r_c \end{cases}$

HENCE

$$U_{ex} = \langle U_N \rangle = \frac{1}{2} V \rho^2 \int g(r)\, v_2(r)\, 4\pi r^2\, dr$$

U_{ex}^c THIS IS WHAT WE HAVE FROM SIMULATIONS, I.E. $\frac{1}{2}\sum_{i,j}{}' v_2^c(r_{ij})$

$$= \frac{1}{2} V \rho^2 \int_0^{r_c} g(r)\, v_2(r)\, 4\pi r^2\, dr +$$

$$+ \frac{1}{2} V \rho^2 \int_{r_c}^{\infty} g(r)\, v_2(r)\, 4\pi r^2\, dr$$

TAIL CORRECTION

Q. CAN WE ESTIMATE SOMEHOW THE TAIL CORRECTION?

$$U_{ex}^{TAIL} = \frac{1}{2} V \rho^2 \int_{r_c}^{\infty} g(r)\, u_2(r)\, 4\pi r^2\, dr$$

↑ TAIL CORRECTION

IF r_c IS BIG ENOUGH FOR $r \geq r_c$ $g(r) \approx 1$

HENCE

THIS INTEGRAL CAN BE CALCULATED ↓

$$U_{ex}^{TAIL} = \frac{1}{2} V \rho^2 \int_{r_c}^{\infty} u_2(r)\, 4\pi r^2\, dr$$

EXAMPLE LENNARD-JONES SYSTEM

$$u_2(r) = u_{LJ}(r) = A\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$$

SO THAT

$$U_{ex}^{TAIL} = \frac{1}{2} V \rho^2 \int_{r_c}^{\infty} 4\pi r^2 A\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right] =$$

$$= V \frac{8\pi}{3} \varepsilon \rho^2 \sigma^3 \left[\frac{1}{3}\left(\frac{\sigma}{r_c}\right)^9 - \left(\frac{\sigma}{r_c}\right)^3\right]$$

WE CAN DO ALSO BETTER

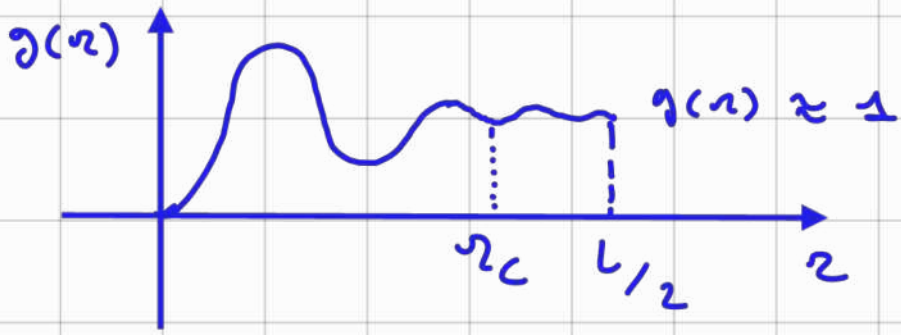
ENERGY FROM SIMULATION

$$U_{ex} = \underbrace{\frac{1}{2} V \rho^2 \int_0^{r_c} g(r)\, u_2(r)\, 4\pi r^2\, dr}_{\text{TAIL CONNECTION}} + \underbrace{\frac{1}{2} V \rho^2 \int_{r_c}^{L/2} g(r)\, u_2(r)\, 4\pi r^2\, dr}_{\approx 1} + \underbrace{\frac{1}{2} V \rho^2 \int_{L/2}^{+\infty} g(r)\, u_2(r)\, 4\pi r^2\, dr}_{\text{TAIL CORRECTION}} =$$

$$= U_{ex}^{C} + U_{ex}^{TAIL}$$

WHERE $g(r)$ CALCULATED FROM SIMULATIONS

$$U_{ex}^{TAIL} = \frac{1}{2} V \rho^2 \int_{r_c}^{L/2} g(r)\, u_L(r)\, 4\pi r^2\, dr + \frac{1}{2} V \rho^2 \int_{L/2}^{+\infty} u_L(r)\, 4\pi r^2\, dr$$



CALCULATION OF PRESSURE FROM $g(r)$

$$P = -\frac{\partial F}{\partial V} = k_B T \frac{\partial \ln Z}{\partial V} \qquad F = -k_B T \ln Q \quad \text{BUT}$$

$$Z = Z_{ID} \cdot Z_{ex} \qquad Z_{ID} = V^N \qquad Q = \frac{\Lambda^{-3N}}{N!} Z$$

$$Z_{ex} = \int e^{-\beta U_N(\vec{r}_1 \ldots \vec{r}_N)} d^N r$$

WHERE $\Lambda = \sqrt{\frac{2\pi\hbar^2}{m k_B T}}$

IT DOES NOT DEPEND ON V

SCALED COORDINATES:

$$\vec{r}_i = V^{1/3} \vec{s}_i$$

$$P = k_B T \frac{\partial}{\partial V} \left(\ln Z_{ID} + \ln Z_{ex} \right) = k_B T \frac{\partial \ln V^N}{\partial V} + k_B T \frac{\partial \ln Z_{ex}}{\partial V} = \underbrace{k_B T \rho}_{P_{ID}} + \underbrace{k_B T \frac{\partial \ln Z_{ex}}{\partial V}}_{P_{ex}}$$

$$P = P_{ID} + P_{ex}$$

$U_N = U_N(V^{1/3} s_1 \ldots V^{1/3} s_N)$

$$P_{ex} = k_B T \frac{1}{Z_{ex}} \frac{\partial Z_{ex}}{\partial V} = \frac{k_B T}{Z_{ex}} \frac{\partial}{\partial V} \left(\int e^{-\beta U_N} d^N s \right) =$$

$$= -\cancel{k_B T}\,\beta \int \frac{\partial U_N}{\partial V}\,\frac{e^{-\beta U_N} V^N}{Z_{ex}\, V^N}\, d\vec{s}^N = -\int \frac{\partial U_N}{\partial V}\,\frac{e^{-\beta U_N}}{Z}\, d\vec{r}^N =$$

$Z_{ex} V^N = Z$

$$= -\left\langle \frac{\partial U_N}{\partial V} \right\rangle$$

$$P = P_{ID} - \underbrace{\left\langle \frac{\partial U_N}{\partial V} \right\rangle}_{P_{EX}}$$

(*) NOTE THAT WE CAN BRING $\frac{\partial}{\partial V}$ INSIDE THE INTEGRAL SINCE THE INTEGRATION DOMAIN DOES NOT DEPEND ON V IF WE USE SCALE COORDS.

$$U_N = \frac{1}{2}\sum_{i,j}{}' U_2(\vec{r}_{ij}) \qquad V = L^3 \Rightarrow \frac{\partial}{\partial V} = \frac{1}{3L^2}\frac{\partial}{\partial L}$$

$$P_{EX} = -\left\langle \frac{1}{3L^2}\frac{\partial U_N}{\partial L} \right\rangle = -\left\langle \frac{1}{3L^2}\frac{\partial}{\partial L}\left(\frac{1}{2}\sum_{ij}{}' U_2(\vec{r}_{ij})\right)\right\rangle =$$

$\vec{s}_{ij} = \frac{\vec{r}_{ij}}{L}$

$$= -\left\langle \frac{1}{6L^2}\sum_{ij}{}' \underbrace{\frac{\partial U_2}{\partial \vec{r}_{ij}}}_{-\vec{f}_{ij}} \cdot \frac{\partial \vec{r}_{ij}}{\partial L} \right\rangle = \left\langle \frac{1}{6L^3}\sum_{ij}{}' \vec{f}_{ij}\cdot\vec{r}_{ij} \right\rangle =$$

NOTE THAT FOR A SYSTEM WITH PBC WE CAN RESTRICT THE CALCULATION TO "FIRST" BOX, SO THAT:

$$U_N = \frac{1}{2}\sum_{\vec{n}}\sum_{i,j} U_2(\vec{r}_i - \vec{r}_j + L\vec{n})$$

WHERE $\vec{n} = (n_x, n_y, n_z)$

WITH n_α INTEGERS

HENCE

$$\frac{\partial U_N}{\partial L} = \frac{1}{2}\sum_{ij}{}' \frac{\partial U_2(\vec{r}_i - (\vec{r}_j + L\vec{n}))}{\partial L} =$$

$$= \frac{1}{2}\sum_{ij}{}' \frac{\partial U_2}{\partial \vec{r}_{ij}} \cdot \frac{\partial(\vec{r}_i - (\vec{r}_j + L\vec{n}))}{\partial L} =$$

$$= \frac{r_i - (r_j + L\vec{n})}{L} = \frac{\vec{r}_{ij}}{L}$$

WHERE j RUNS OVER ALSO PERIODIC PARTICLES

(FORCE ACTING BETWEEN I AND J)

$$= \frac{1}{6V} \left\langle \sum_{ij}{}' \vec{f}_{ij} \cdot \vec{r}_{ij} \right\rangle$$

SIMILAR EXPRESSION FOR MOLECULAR PRESSURE

$$P = \rho k_B T + \underbrace{\frac{1}{6V} \left\langle \sum_{ij}{}' \vec{f}_{ij} \cdot \vec{r}_{ij} \right\rangle}_{P_{EX}}$$

WE CAN NOW WRITE P_{EX} IN TERMS OF $g(\vec{r})$

ALL TERMS OF $\sum_{ij}{}'$ ARE EQUAL

$$P_{EX} = \frac{N(N-1)}{6V} \int \underbrace{\frac{e^{-\beta U_N}}{Z}} \vec{f}_{12} \cdot \vec{r}_{12} \, d\vec{r}_1 d\vec{r}_2 \underline{d\vec{r}^{(N-2)}} =$$

$$= \frac{1}{6V} \int \underline{\rho^{(2)}(\vec{r}_1, \vec{r}_2)} \vec{f}_{12} \cdot \vec{r}_{12} \, d\vec{r}_1 d\vec{r}_2$$

CHANGE OF VARIABLES

$(\vec{r}_1, \vec{r}_2) \rightarrow (\vec{r}, \vec{r}_2)$ WHERE $\vec{r} = \vec{r}_1 - \vec{r}_2$

$$= \frac{1}{6V} \int \rho^{(2)}(\vec{r} + \vec{r}_2, \vec{r}_2) \vec{f}(\vec{r}) \cdot \vec{r} \, d\vec{r} \, d\vec{r}_2 =$$

IF SYSTEM IS HOMOGENEOUS $\rho^{(1)}(\vec{r} + \vec{r}_2, \vec{r}_2) = \rho^{(2)}(\vec{r})$ HENCE

$$= \frac{1}{6} \int \rho^{(2)}(\vec{r}) \, \vec{f}(\vec{r}) \cdot \vec{r} \, d\vec{r}$$

BUT $\rho^{(2)}(\vec{r}) = \rho^2 g(\vec{r})$

$$= \frac{\rho^2}{6} \int \vec{f}(\vec{r}) \cdot \vec{r} \, g(\vec{r}) \, d\vec{r}$$

IF THE SYSTEM IS ISOTROPIC

$$P_{EX} = \frac{\rho^2}{6} \int f(r) \, g(r) \, 4\pi r^3 \, dr$$

TAIL CORRECTIONS P_{SIM} FROM SIMULATIONS

$$P_{EX} = \underbrace{\frac{\rho^2}{6}\int_0^{r_c} f(r)\, g(r)\, 4\pi r^3 dr}_{P_{SIM}} + \underbrace{\frac{\rho^2}{6}\int_{r_c}^{+\infty} f(r)\, \cancel{g(r)}^{\approx 1}\, 4\pi r^3 dr}_{\text{FOR LJ WE CAN CALCULATE THIS ANALITICALLY}} = P_{TAIL}$$

(•) NOTE THAT

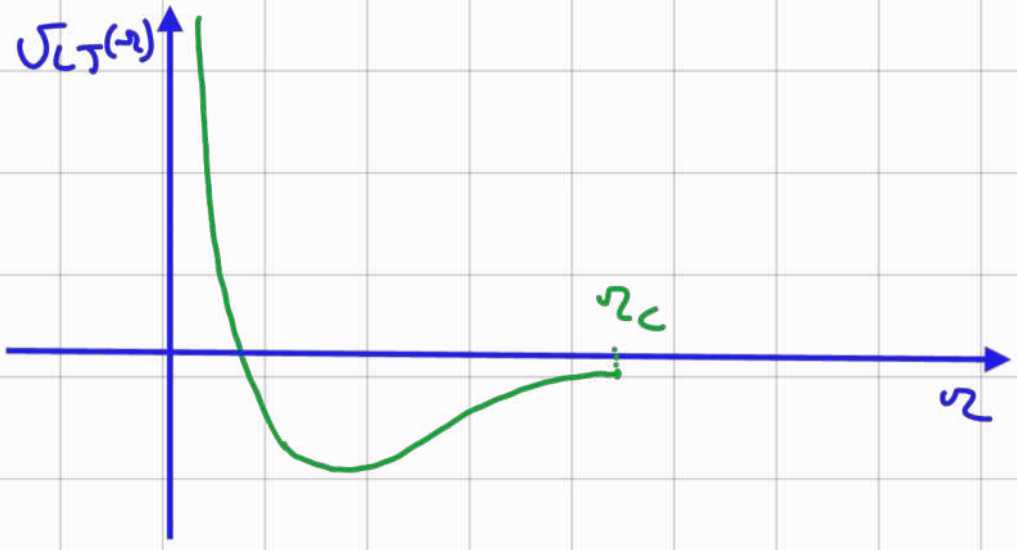
WE CAN CALCULATE THIS FROM SIMULATIONS

$$P_{SIM} = \frac{\rho^2}{6}\int_0^{r_c} f(r)\, g(r)\, 4\pi r^3 dr = \frac{1}{6V}\left\langle \sum_{ij}{}' \vec{F}_{ij}^{c} \cdot \vec{r}_{ij} \right\rangle$$

THE ABOVE TAIL CORRECTION APPLIES IF WE WANT TO COMPARE OUR RESULTS WITH A SYSTEM WITHOUT CUTOFF (•). E.G WE HAVE : $P = P_{SIM} + P_{TAIL}$

Q. BUT WHAT IS THE PRESSURE OF A SYSTEM WITH TRUNCATED INTERACTIONS?

FOR EXAMPLE CONSIDER A TRUNCATED LJ SYSTEM



AROUND r_c ONE HAS

$$v_{LT}(r) \approx \theta(r_c - r)\, v_{LT}(r_c)$$

WHERE

$$\theta(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$f(r) = -\frac{dv_2}{dr} = v_{LT}(r_c)\,\delta(r - r_c)$$

SINCE

$$\frac{d}{dx}\theta(x) = \delta(x)$$

HENCE WE HAVE

$$P_{IMP} = \frac{1}{6}\rho^2 \int_{r_c-\varepsilon}^{\infty} v_{LT}(r_c)\,\delta(r - r_c)\,4\pi r^3\,dr =$$

$$= \frac{1}{6}\rho^2\, v_{LT}(r_c)\,4\pi r_c^3$$

"IMPULSIVE" CORRECTION TO PRESSURE

STATIC STRUCTURE FACTOR

1 POINT DENSITY

$$\rho(\vec{r}) = \left\langle \sum_i \delta(\vec{r} - \vec{r}_i) \right\rangle$$

THEN

$$\langle \rho(\vec{k}) \rangle = \underbrace{\int \rho(\vec{r}) e^{-i\vec{k}\cdot\vec{r}} \, d\vec{r}}_{\text{FOURIER TRANSFORM}} = \left\langle \underbrace{\sum_i e^{-i\vec{k}\cdot\vec{r}_i}}_{\rho(\vec{k})} \right\rangle$$

↑ ENSEMBLE AVERAGE

WHERE $\rho(\vec{k}) = \sum_i e^{-i\vec{k}\cdot\vec{r}_i}$

WE DEFINE

$$S(\vec{k}) \equiv \frac{1}{N} \langle \rho(\vec{k}) \rho^*(\vec{k}) \rangle$$

↑ ENSEMBLE AVERAGE

← $S(\vec{k})$ CAN BE MEASURED EXPERIMENTALLY THROUGH SCATTERING EXPERIMENTS

BUT

$$S(\vec{k}) = \frac{1}{N} \left\langle \sum_{i,j} e^{-ik(\vec{r}_i - \vec{r}_j)} \right\rangle = 1 + \frac{1}{N} \left\langle \sum_{i,j}' e^{-i\vec{k}\cdot(\vec{r}_i - \vec{r}_j)} \right\rangle =$$

($\sum'$: $i \neq j$)

$$= 1 + \frac{1}{N} \left\langle \int \sum_{i,j}' e^{-ik\cdot(r \rightleftarrows r')} \delta(\vec{r} - \vec{r}_i)\, \delta(\vec{r}' - \vec{r}_j) \, d\vec{r}\, d\vec{r}' \right\rangle =$$

$$= 1 + \frac{1}{N} \int d\vec{r}\, d\vec{r}' \, e^{-i\vec{k}\cdot(\vec{r} - \vec{r}')} \underbrace{\left\langle \sum_{i,j}' \delta(\vec{r} - \vec{r}_i)\, \delta(\vec{r} - \vec{r}_j) \right\rangle}_{\rho^{(2)}(\vec{r}, \vec{r}')} =$$

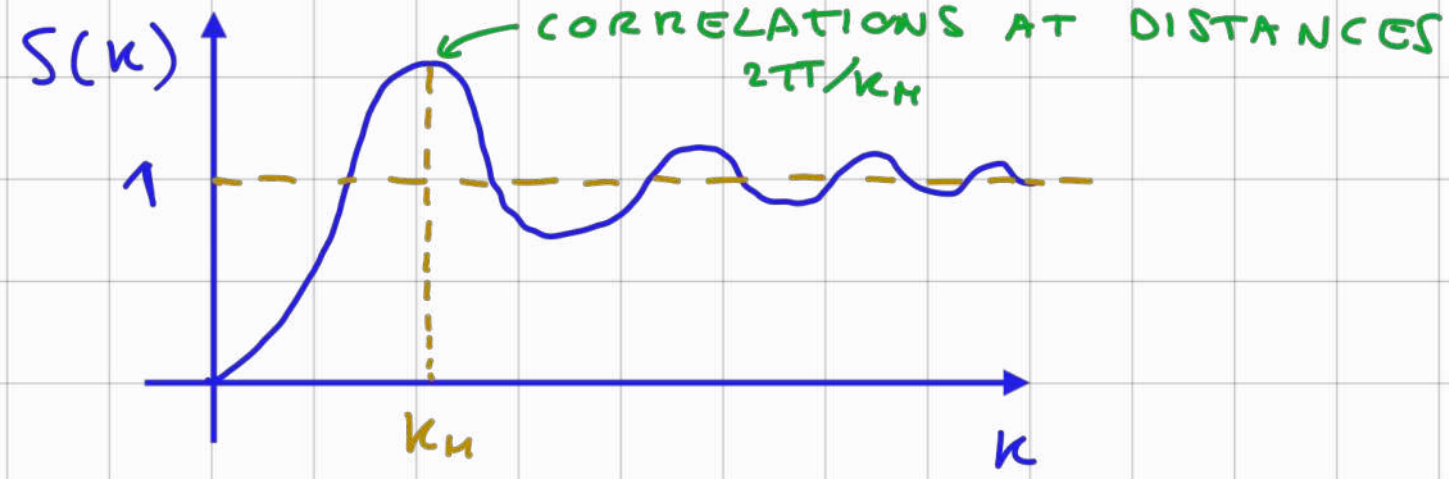
$$= 1 + \frac{1}{N} \int d\vec{r}\, d\vec{r}' \, e^{-ik(\vec{r} - \vec{r}')} \underbrace{\rho^2 g^{(2)}(\vec{r}, \vec{r}')}_{\rho^{(2)}(\vec{r}, \vec{r}')} =$$

ASSUMING THE SYSTEM HOMOGENEOUS

$$= 1 + \frac{\rho^2}{N} V \int d\vec{r} \, e^{-i\vec{k}\cdot\vec{r}} \, g(\vec{r}) =$$

$$= 1 + \rho \underbrace{\int d\vec{r} \, e^{-i\vec{k}\cdot\vec{r}} \, g(\vec{r})}_{\text{FOURIER TRANSFORM OF } g(\vec{r})}$$

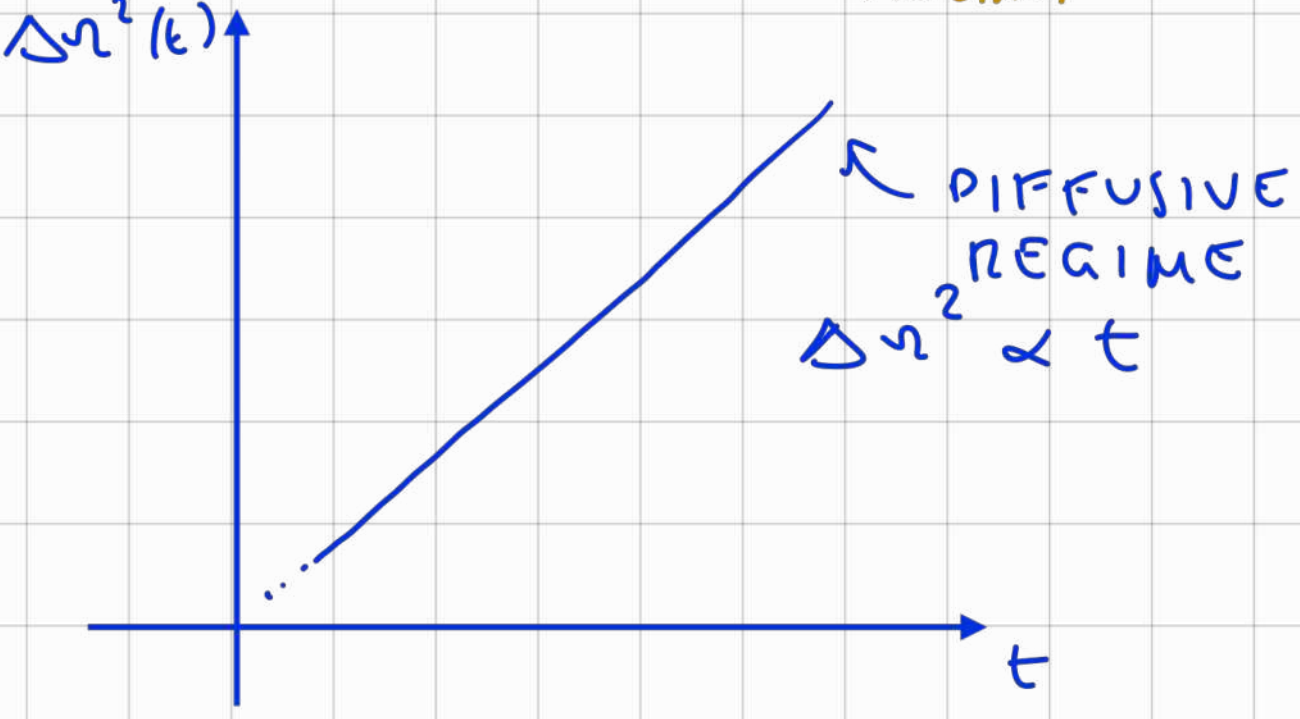
IN THE ISOTROPIC CASE



MEAN SQUARED DISPLACEMENT

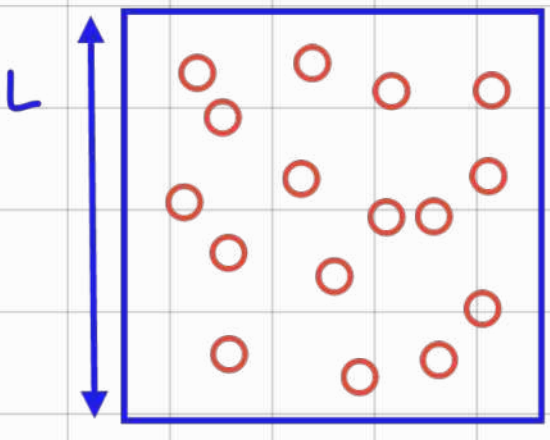
$$\Delta r^2(t) \equiv \frac{1}{N} \sum_{i=1}^{N} \| \vec{r}_i(t) - \vec{r}_i(0) \|^2$$

- $\Delta r^2(t)$: FUNCTION OF t
- $\vec{r}_i(t)$: PASSO MC
- $\vec{r}_i(0)$: POSIZIONI INIZIALI



DIFFUSIVE REGIME

$\Delta r^2 \propto t$



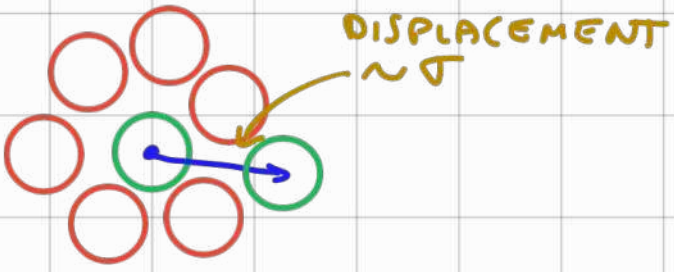
$\sqrt{\Delta r^2} \gtrsim L$

THE SYSTEM CAN BE CONSIDERED EQUILIBRATED SINCE ALL PARTICLES HAVE MOVED ENOUGH I.E. HAVE FULLY EXPLORED THE CONFIGURATION SPACE

WE COULD ALSO USE A LESS STRINGENT CONDITION, I.E.

TYPICAL PARTICLE SIZE

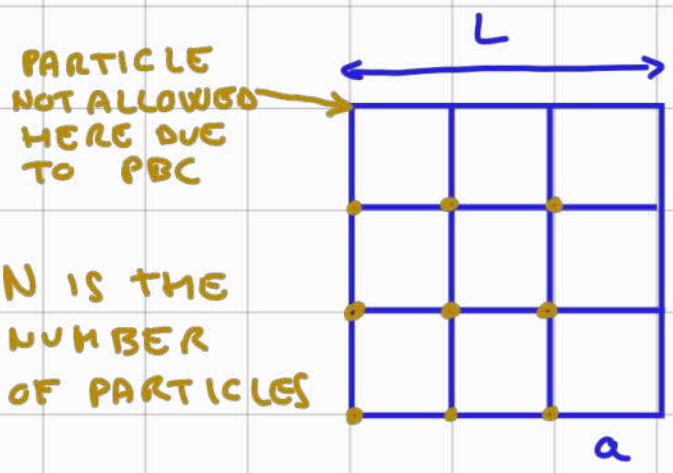
$$\sqrt{\Delta r^2} \gtrsim \sigma$$



MAIN CONTRIBUTION TO SPATIAL CORRELATIONS COMES FROM FIRST PEAK OF $S(R)$ WHICH ROUGHLY CORRESPONDS TO A DISTANCE EQUAL TO σ

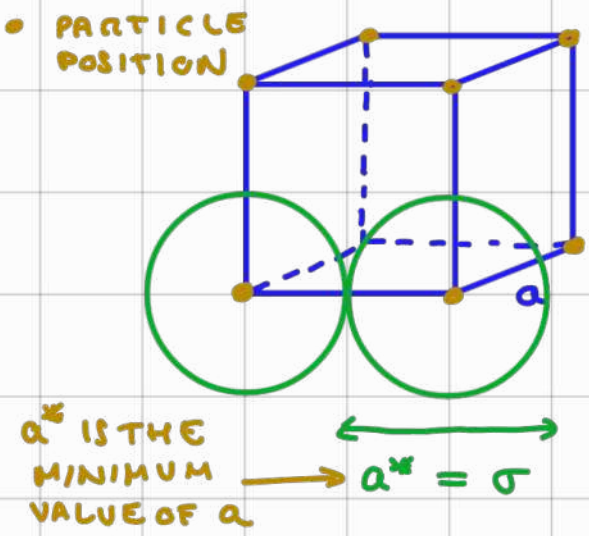
INITIAL CONFIGURATION

CUBIC LATTICE



N IS THE NUMBER OF PARTICLES

EACH PARTICLE HAS DIAMETER σ



a^* IS THE MINIMUM VALUE OF a → $a^* = \sigma$

$$\phi = \frac{N v_0}{L^3} = \frac{n \left(\frac{L}{a}\right)^3 v_0}{L^3}$$

WHERE v_0 IS THE VOLUME OF A PARTICLE AND n IS THE NUMBER OF PARTICLES PER UNIT CELL ($n=1$ IN THIS CASE)

$$= n \frac{v_0}{a^3} = \frac{n v_0}{a^3}$$

BUT $v_0 = \frac{\pi}{6}\sigma^3$ IF PARTICLE IS SPHERICAL

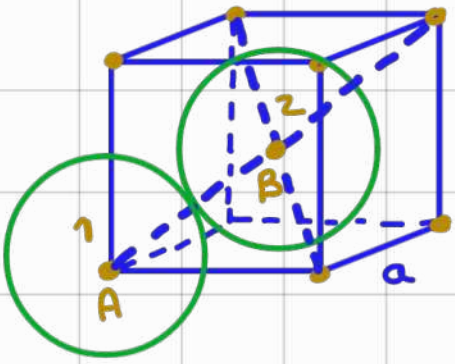
HENCE CONSIDERING THAT $a^* = \sigma$

$$\phi^* = \{\text{MAXIMUM } \phi\} = \frac{v_0}{(a^*)^3} = \frac{\pi}{6}\frac{\sigma^3}{\sigma^3} = \frac{\pi}{6}$$

IF WE WANT A LOWER $\phi = \phi'$ IT IS JUST A MATTER TO EXPAND THE BOX AND SCALE PARTICLE POSITIONS BY A FACTOR ξ SUCH THAT $L' = \xi L$ $\vec{r}_i' = \xi \vec{r}_i$

WITH $\xi = \left(\frac{\phi^*}{\phi'}\right)^{1/3}$ I.E. $\phi' = \frac{\phi^*}{\xi^3}$

HIGHER ϕ: BODY CENTERED CUBIC (BCC)



$$\phi^* = \frac{n\, v_0}{(a^*)^3}$$

WHERE IN THIS CASE

$n = 2$ AND

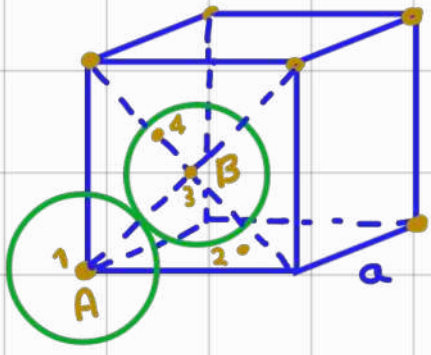
a^* IS SUCH THAT

$$\overline{AB} = \sigma = \frac{\sqrt{3}\, a^*}{2} \quad \text{I.E. } a^* = \frac{2}{\sqrt{3}}\sigma$$

SO THAT

$$\phi^* = \frac{2\,\frac{\pi}{6}\,\sigma^3}{\left(\frac{2}{\sqrt{3}}\right)^3 \sigma^3} = \frac{\pi}{3}\left(\frac{\sqrt{3}}{2}\right)^3 \approx 0.68$$

EVEN HIGHER ϕ : FACE CENTERED CUBIC (FCC)



$$\phi^* = \frac{n\, v_0}{(a^*)^3}$$

WHERE IN THIS CASE

$n = 4$ AND

a^* IS SUCH THAT

$$\overline{AB} = \sigma = \frac{\sqrt{2}}{2}\, a^* \quad \text{I.E. } a^* = \frac{2}{\sqrt{2}}\sigma$$

SO THAT

$$\phi^* = \frac{4\,\frac{\pi}{6}\,\sigma^3}{\left(\frac{2}{\sqrt{2}}\right)^3 \sigma^3} = 2\left(\frac{\sqrt{2}}{2}\right)^3 \frac{\pi}{3} \approx 0.74$$

LINKED CELL LISTS

IN A METROPOLIS MC A MC STEP CONSISTS OF ATTEMPTING TO MOVE N PARTICLE

FOR EACH PARTICLE

$$acc(o \to n) = \min\{1, e^{-\beta \Delta U}\}$$

$\Delta U = U_i' - U_i$ WHERE

$U_i = \sum_j u_2(\vec{r}_{ij})$ WHICH REQUIRES N CALCULATIONS OF u_2 (NOTE THAT EVEN IF $r > r_c$ WE NEED TO CALCULATE $|\vec{r}|$)

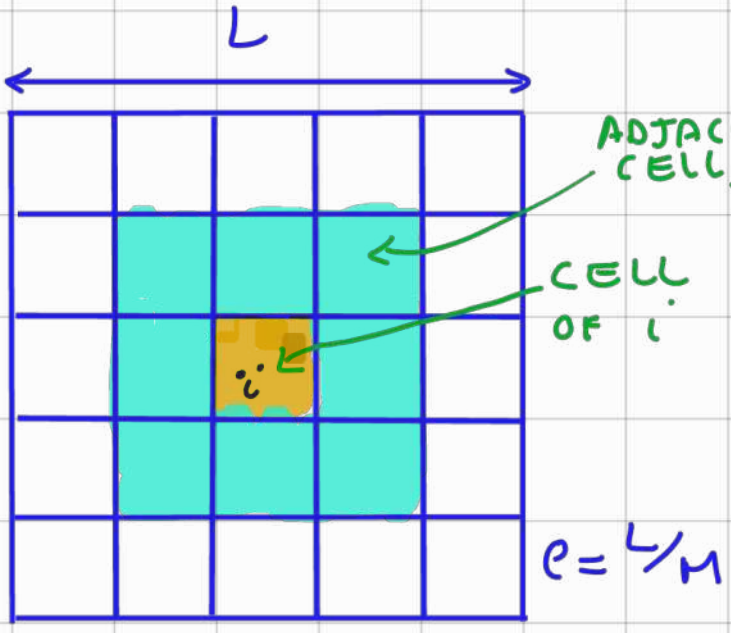
HENCE FOR A MC STEP COMPLEXITY IS $O(N \times N) = O(N^2)$

WE WANT TO REDUCE THE NUMBER OF TIMES WE HAVE TO CALCULATE u_2

LET'S CONSIDER M^3 CUBIC CELLS SUCH THAT $L/M \geq r_c$ ← CUTOFF RADIUS

I.E. M IS THE LARGEST INTEGER SUCH THAT

$$\ell = \frac{L}{M} > r_c$$



TO CALCULATE U_i AND U_i' WE CAN JUST CONSIDER PARTICLES IN THE 27 CELLS (I.E. THE CELL TO WHICH PARTICLE i BELONGS AND THE 26 ADJACENT CELLS)

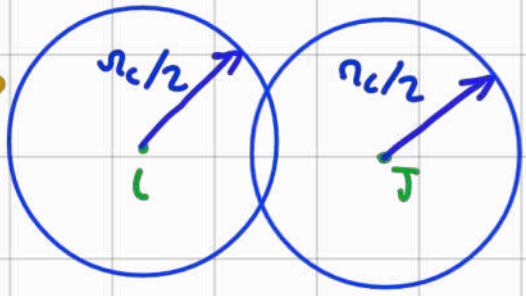
WITH THIS STRATEGY FOR EACH MC MOVE WE HAVE TO CALCULATE $\tilde{N}$ TIMES U_L WITH

$$\tilde{N} \approx N \, 27 \cdot r_c^3 \rho = N \underbrace{27 r_c^3 \rho}_{\text{THIS IS A CONSTANT IF WE INCREASE BOX SIZE KEEPING FIXED } \rho} = O(N)$$

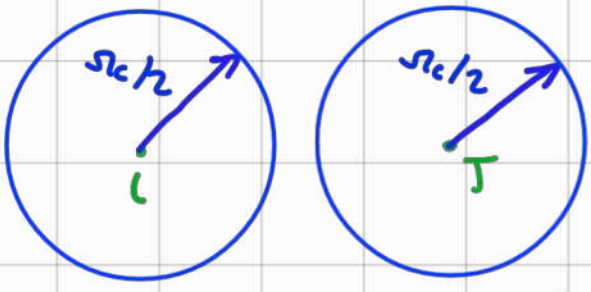
NEAREST NEIGHBOUR LISTS (NNL)

CONSIDER TWO PARTICLES i AND j AND OBSERVE THAT:

SPHERE OF RADIUS r_c (CUTOFF RADIUS)



THE TWO SPHERES OVERLAP HENCE I AND J INTERACT

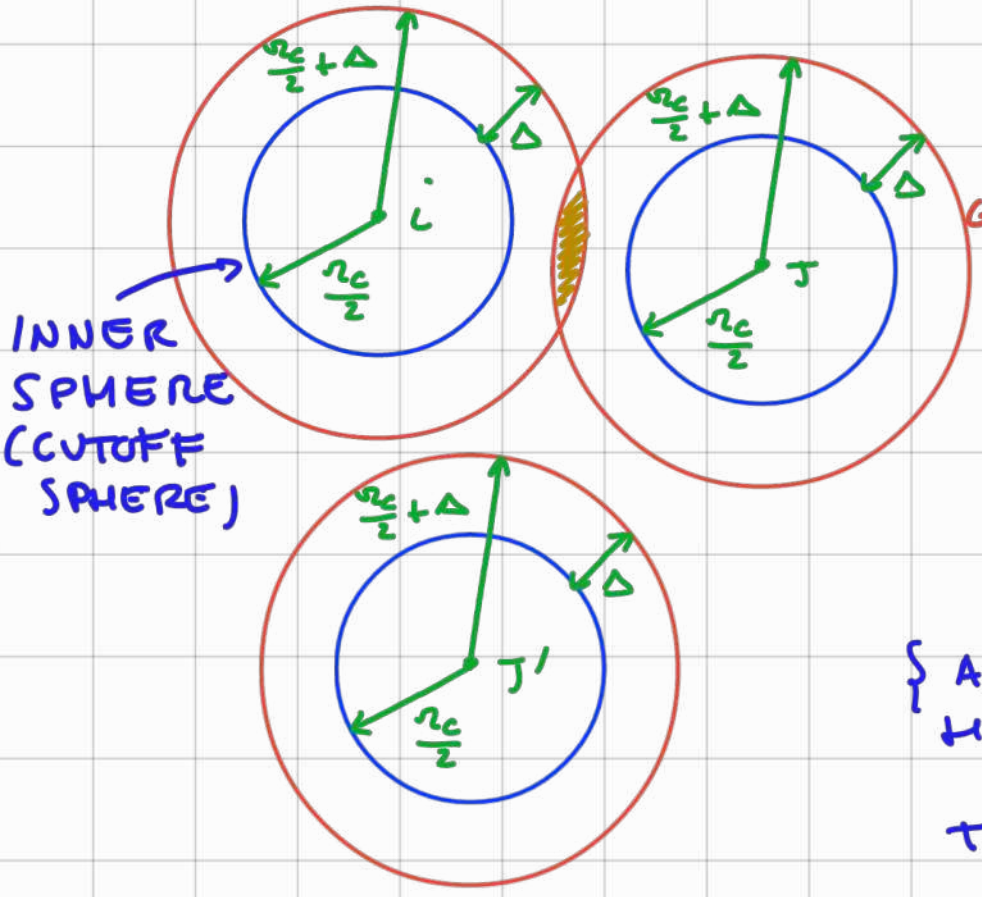


THE TWO SPHERES DO NOT OVERLAP HENCE i AND j DO NOT INTERACT

AT A CERTAIN TIME t_0 (MC STEP) WE BUILD THE NNL OF ALL PARTICLES (STARTING FROM FIRST SIMULATION STEP)

CONSIDER THE BUILDING OF NNL OF A PARTICLE i

j AND j' ARE TWO PARTICLES OTHER THAN i



INNER SPHERE (CUTOFF SPHERE)

OUTER SPHERE

$NNL(i) =$

{ LIST OF ALL PARTICLES BELONGING TO NNL OF i } =

{ ALL PARTICLES WHICH HAVE OUTER SPHERES OVERLAPPING WITH THAT OF i }

HENCE IN THIS CASE:

$j \in NNL(i)$

SINCE SPHERES OF RADIUS $\frac{r_c}{2}+\Delta$ OVERLAP

$j' \notin NNL(i)$

SINCE SPHERES OF RADIUS $\frac{r_c}{2}+\Delta$ DO NOT OVERLAP

FOR $t > t_0$ INNER SPHERE ARE PLACED WITH THEIR CENTER AT PARTICLE POSITION WHILE OUTER SPHERES KEEP FIXED

AS LONG AS EACH INNER SPHERE REMAIN INSIDE ITS CORRESPONDING OUTER SPHERE WE CAN CALCULATE U_i AND U_i' BY CONSIDERING $\forall i$ THE PARTICLES $\in NNL(i)$

SINCE IF TWO INNER SPHERES ARE INSIDE TWO NON OVERLAPPING OUTER SPHERES THEY CAN NOT OVERLAP (AND CORRESPONDING PARTICLES CAN NOT INTERACT)

AN INNER SPHERE ESCAPE IF $\| \vec{r}_i(t) - \vec{r}_i(t_0) \| \geq \Delta$

IF THIS HAPPENS ALL NNL MUST BE REBUILT

Δ IS A PARAMETER WHICH MUST BE OPTIMIZED



AND IF η_{LL} IS THE EFFICIENCY OF LL

$$\frac{n_{NNL}}{n_{LL}} = \frac{\{\#\text{ PARTICLES IN THE 27 LL CELLS}\}}{\{\#\text{ PARTICLES IN THE NNL(i)}\}}$$

↑ SINCE FOR EACH PARTICLE WE NEED TO TEST IF $r < r_c$

$$\frac{\rho\, r_c^3\, 27}{\frac{4\pi}{3} r_c^3\, \rho} \approx \frac{27 \cdot 3}{4\pi} \approx 6$$

↑ IF $\Delta \ll r_c$